

Predicting Contract Renewal in Public Digital Procurement

A Survival-Analysis Approach on BOAMP Notices for Gigalis
Internship Report — Executive Orientation

Internship Report — Gigalis (Pays de la Loire)

June 2026

Abstract

This is the **executive report** of the internship: it lets a reader scan the project rapidly and understand *what was done*, *why it matters*, and *what is reliable*. The full implementation and diagnostics are in the **technical report** (`phase1_technical_report.pdf`); the data reliability and event-definition detail are in the **data quality report**.

The work concerns predictive analysis of public digital procurement for **Gigalis**, a central purchasing body in Pays de la Loire. The goal is to move from a descriptive view (“what happened?”) towards a predictive one (“when will a contract be re-tendered?”). Because BOAMP contains *no explicit recurrence-chain field*, the relationship is reconstructed from observable signals: a proxy recurrence **event** built from Sentence-Transformer text embeddings combined with CPV, temporal, and buyer signals, used as the event variable for survival analysis. A method-comparison experiment tested the calibrated composite rule (M0) against probabilistic linkage (M1) and active-learning-assisted linkage (M2), with the synthetic benchmark scored by the same Sentence-Transformer encoder as the real pipeline. The current selected method is **M2 balanced** (match-probability threshold 0.65): on 1,210 eligible source contracts it records **254 proxy recurrence events (21.0%)**, improving benchmark-estimated precision (0.612 vs. 0.575) and recall (0.733 vs. 0.568) over M0 balanced and lowering the negative-control acceptance rate (7.9% vs. 9.4%). **M0 balanced** (text ≥ 0.50 , composite ≥ 0.50 , $W = 6$, corrected generic CPV handling; 269 events, 22.2%) is retained as the conservative transparent baseline, and the older 665-event output only as the pre-calibration baseline. Mapped manual-audit precision was excluded from the selection because the audit sample was drawn from the pre-calibration baseline’s links (a baseline-anchored diagnostic). Under M2 balanced, the Kaplan–Meier median is not reached, the official reduced-spec Cox C-index is **0.553** (PH-checked), and the best AFT model is LogNormalAFT (AIC 3,357.9). Operational 12- and 24-month indicators are available for 1,204 contracts, with median probabilities 0.0206 and 0.0634. Headline survival results are stable across M0 and M2 balanced. The headline caveat: **event** is a proxy recurrence outcome, not BOAMP ground truth; real BOAMP precision and recall are not directly observable.

Methodological Note — From Maximizing Linkage to Maximizing Reliability (2026-07-05)

This note explains *why* the event definition changed after the main pipeline was first built, not just *what* changed. It is the key to reading the rest of this report: numbers in Section 4 were computed under the **initial** linking method; the numbers in this note and in Section 5 reflect the **calibrated** method that should be used for any downstream decision.

Initial method and its results. The first version of the pipeline (Section 3.3) linked a source contract to *any* later same-buyer notice whose composite score was non-negative after passing a soft text-similarity filter. This maximized the linking rate: 665 of 1,210 eligible contracts (55.0%) were linked, against a 7.6% lexical baseline — a large apparent improvement.

Challenge discovered. BOAMP has no official field certifying that one notice renews another, so this 55.0% figure could not be checked against ground truth directly. A manual audit of 150 stratified cases against the full official BOAMP record (2026-07-02) showed the initial rule was too permissive: raw precision was only ≈ 0.15 (weighted ≈ 0.09 over the 665-event population), rising to 0.50 in the high-confidence tier and 1.00 only above text similarity 0.80; 12.8% of decided-unlinked sources also hid a plausible missed renewal. In short, the initial rule optimized for *linking rate*, not for *link correctness*, and there is no legal ground truth against which to prove any individual link is right.

Reason for changing method. Because no certified renewal label exists for BOAMP, the true precision/recall of any rule cannot be measured on the real data alone. The project therefore stopped trying to maximize the number of linked events and instead built a **synthetic BOAMP-like benchmark** with known recurrence truth by construction, so that candidate rules could be compared on measurable precision, recall, and F1 before being re-applied to real BOAMP. The goal changed from “*detect every true renewal*” to “*construct a reliable, calibrated proxy-event definition*.”

New calibrated method. The synthetic benchmark generates synthetic source notices, candidate notices, and an explicit true-link table under three difficulty scenarios (easy, medium, hard), with real-BOAMP-informed design parameters for timing, buyer-name variation, generic or drifting CPV codes, missing duration/CPV, short or paraphrased text, and large-buyer ambiguity. It is scored with the *same Sentence-Transformer encoder as the real pipeline* (paraphrase-multilingual-MiniLM-L12-v2), so benchmark performance is directly comparable to the operational linking setup. A full threshold grid was scored on the benchmark and three explicit rules were selected: **broad**, **balanced**, and **strict**. The calibrated rule family is called **M0** in the later method comparison; its balanced member — text similarity ≥ 0.50 , composite score ≥ 0.50 , temporal window $W = 6$, corrected generic CPV scoring, no margin floor — was the recommended main rule until the 2026-07-08 method comparison and is now the conservative baseline. On the synthetic benchmark, the balanced rule scores precision/recall/F1 of 0.777/0.806/0.791 (easy), 0.601/0.586/0.593 (medium), and 0.318/0.302/0.310 (hard, Figure 1) — it was chosen over broad and strict because it keeps acceptable reliability in the medium (realistic-noise) scenario while still preserving enough real events for survival modeling.

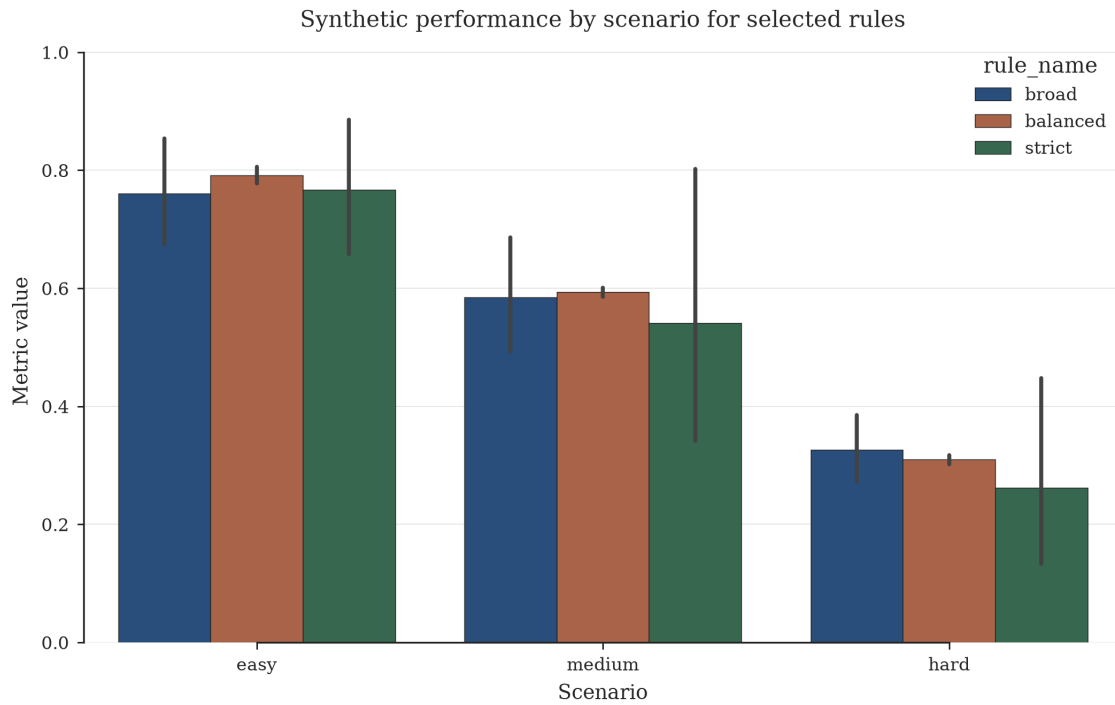


Figure 1: Synthetic-benchmark performance of the broad, balanced, and strict rules across the easy/medium/hard scenarios. All three rules degrade under harder synthetic noise; the balanced rule was chosen for its stability and its yield of real events, not because it is the single most precise option on paper.

Results after the change. Re-applied to real BOAMP, the balanced rule produces **269 proxy recurrence events** among 1,210 eligible source contracts (**22.2%**) — down from 665 (55.0%) under the initial rule. The broad sensitivity gives 490 events (40.5%); the strict sensitivity gives 79 events (6.5%, underpowered for Cox/AFT). Figure 2 shows this drop in scale.

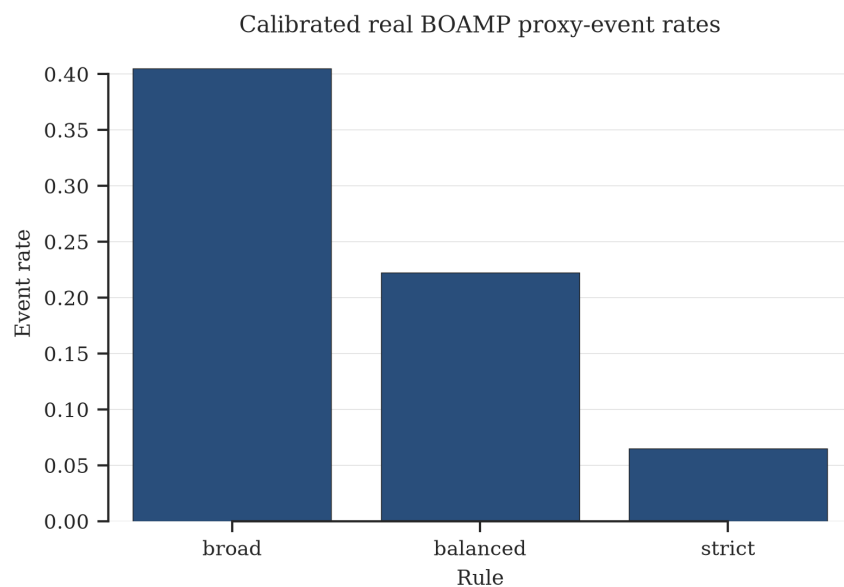


Figure 2: Proxy-event rate on real BOAMP under the three calibrated rules (1,210 eligible contracts). The balanced rule keeps less than half the events of the broad rule, trading linkage quantity for a rule whose behaviour is checked on the synthetic benchmark.

The calibrated survival-ready file is `data/processed/boamp_phase2_survival_calibrated_balanced.csv`,

row-identical to the M0 balanced baseline file `data/processed/boamp_phase2_survival_method_m0_balanced.csv` (the selected main input since 2026-07-08 is `boamp_phase2_survival_method_m2_balanced.csv`; see the method-comparison section below). Its Kaplan–Meier median is not reached; survival is 95.9% at 12 months, 92.3% at 24 months, 83.2% at 48 months, and 75.2% at 60 months (Figure 3). LogNormalAFT is the best balanced-rule AFT model by AIC. Re-scored 12/24-month risk indicators under this rule give a mean 12-month recurrence probability of 0.023 and 27.5 expected recurrences within 12 months (`reports/tables/survival/renewal_risk_12_24_months_calibrated_balanced.csv`).

Two Cox specifications, one dataset. Two Cox models were fitted on this same 269-event balanced data: an initial richer model with category-segment covariates (penalizer 0.1, C-index 0.592, AFT AIC 3,544.8, notebook `07_calibrated_survival_analysis.ipynb`), and a reduced 3-covariate model (`declared_duration_months`, `dur_was_imputed`, `start_year`; penalizer 0.05) used in the 2026-07-08 method-comparison rerun so that M0, M1, M2, and the strict variant are compared on identical covariates (`scripts/linkage_method_comparison_no_ground_truth.py`). Both are valid fits; the reduced spec is the official one because it is the specification used for the final method-comparison decision. For the M0 balanced baseline it gives C-index 0.544 (AFT AIC 3,542.5; richer category-aware spec 0.592), and for the **selected M2 balanced method the official L3 Cox result is the reduced-spec C-index of 0.553** (AFT AIC 3,357.9; richer spec 0.606). The Schoenfeld residual test on the official reduced spec flags a PH violation for `declared_duration_months` ($p < 10^{-24}$, both methods) and passes for `dur_was_imputed` and `start_year` — the same pattern as the pre-calibration model, for which the log-normal AFT above is used as the PH-assumption-free cross-check.

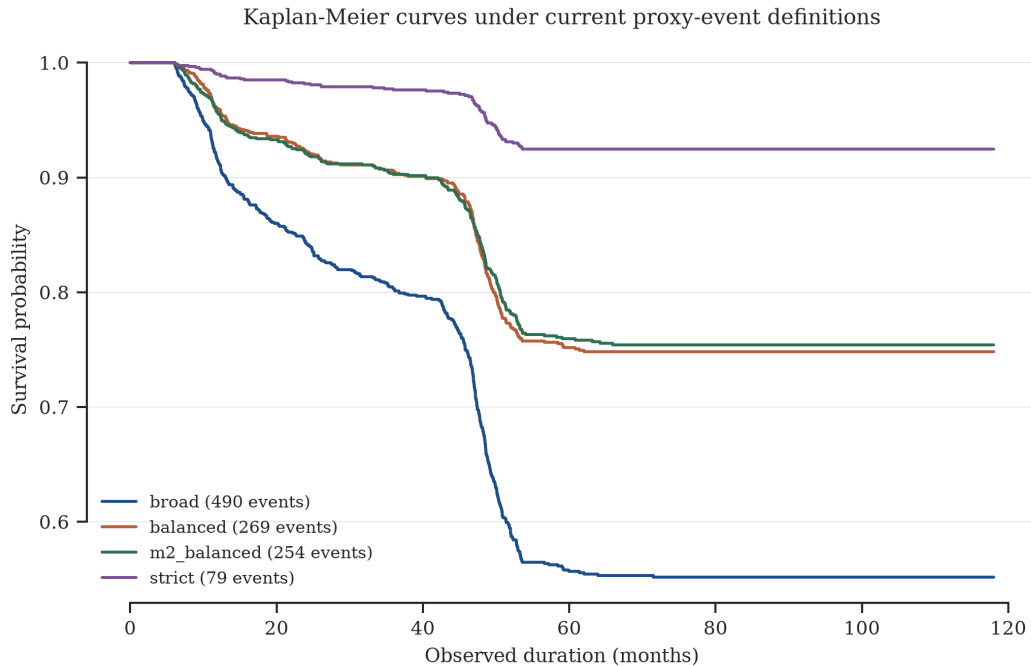


Figure 3: Kaplan–Meier survival curves under the current proxy-event definitions (M0 broad/balanced/strict and the selected M2 balanced). The stricter the rule, the fewer and later the recorded events, and the higher the residual survival at any horizon; the M2 balanced curve tracks the M0 balanced curve closely.

Limitation. Because BOAMP does not provide an official renewal label, the constructed `event` variable — under any of the three rules — should be interpreted as a proxy for *likely procurement recurrence*, not as verified renewal-chain ground truth. The synthetic benchmark quantifies how

the linking method behaves under controlled noise; it does not certify that any specific real link is a verified renewal-chain link. These labels remain proxy recurrences, not externally verified outcomes.

Conclusion of this note. The methodological change is the main deliverable of the calibration phase: the project moved from **maximizing linkage quantity** (665 events, 55.0%, uncalibrated) to **maximizing analytical reliability** (M0 balanced, 269 events, 22.2%, calibrated against a controlled benchmark; then the selected M2 balanced, 254 events, 21.0%, in the method comparison below). The rest of this report documents all stages so the reader can see what changed and why. The detailed implementation is reproducible from `notebooks/04_synthetic_boamp_benchmark.ipynb` and `notebooks/05_parameter_calibration_benchmark_and_real.ipynb`, with generated artifacts tracked under `data/synthetic/` and `reports/tables/validation/`. Section 5 gives the full manual-audit and robustness evidence behind this decision.

Method-Comparison Update — M0, M1, and M2 (2026-07-08)

The latest notebook, `notebooks/06_linkage_method_comparison_no_ground_truth.ipynb`, compares the calibrated rule (M0) with two literature-supported alternatives: a probabilistic linkage classifier (M1) and an active-learning-assisted variant (M2). Splink and recordlinkage were not available in the environment, so M1/M2 were implemented as transparent logistic match-probability models trained on synthetic pair labels, with mapped manual TP/FP labels added for M2 where available. The synthetic pairs are scored with the same Sentence-Transformer encoder as the real pipeline, so all methods are benchmarked on a common footing. This is a comparison under no legal ground truth: real BOAMP precision and recall are not directly observable.

Table 1: Latest real BOAMP method-comparison result. The mapped-audit column is a baseline-anchored diagnostic (the audit sample was drawn from the pre-calibration baseline’s links) and is not a selection criterion.

Method	Variant	Events	Event rate	Neg.-control accept.	Mapped-audit diagnostic
M0	Balanced	269	22.2%	9.4%	43.8% ($n=32$)
M1	Balanced	256	21.2%	8.5%	37.1% ($n=35$)
M2	Balanced	254	21.0%	7.9%	37.1% ($n=35$)
M0	Strict	79	6.5%	0.4%	85.7% ($n=14$)

Across all synthetic scenarios, M0 balanced scores benchmark-estimated precision 0.575 and recall 0.568; M1/M2 balanced score 0.612 and 0.733, and improve on M0 balanced in every scenario (easy, medium, hard) and on the generic-CPV subset. The final recommendation **promotes M2 balanced to the main method**: it passes all five promotion criteria of the transparent, audit-free selection rule (selection score 0.797 vs. 0.741; negative controls 7.9% vs. 9.4%; 254 events; smaller synthetic-to-real accepted-link profile shift, 0.045 vs. 0.128; same text backend as the real pipeline). **M0 balanced is retained as the conservative transparent baseline** sensitivity. Mapped audit precision was excluded from the decision because the 150-case audit sample was stratified on the pre-calibration baseline’s own links and confidence tiers; with only 32–35 decided overlapping pairs per method, its differences are also within sampling noise. M1 — which never touches the audit labels — shows nearly identical gains, so the promotion does not depend on the audit-informed training. The active-learning output is a 150-pair review queue: 25 pairs in each of six review categories (near-threshold, high-score generic CPV, event=0 with high nearest candidate, low top1–top2 margin, high text with CPV mismatch, and same-buyer/same-CPV weak-text); a method-neutral audit of this queue is the recommended next validation step.

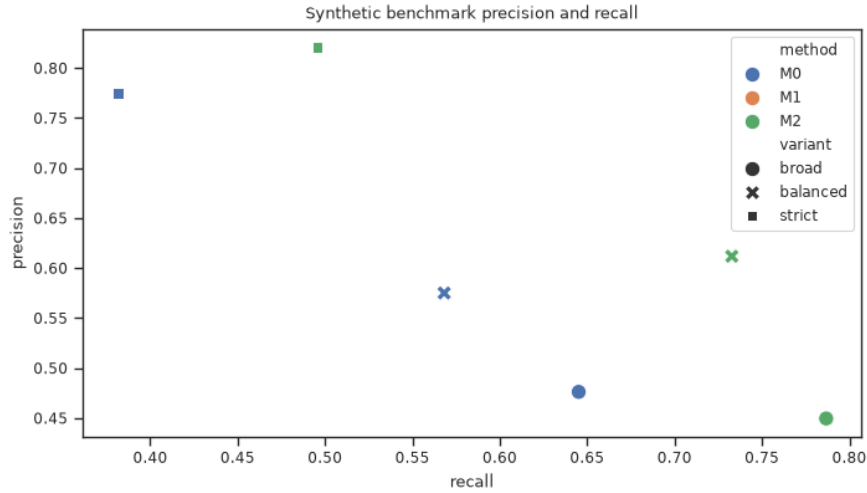


Figure 4: Synthetic benchmark-estimated precision and recall for M0, M1, and M2. The synthetic benchmark supports comparing methods, but it does not prove real BOAMP truth.

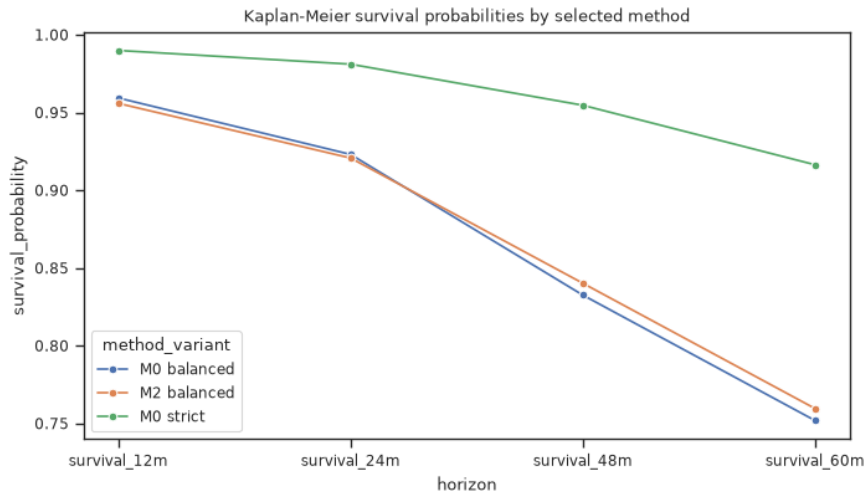


Figure 5: Fixed-horizon Kaplan–Meier survival probabilities for M0 balanced, M2 balanced, and M0 strict. M2 balanced is the selected main method; its survival profile is close to the M0 balanced baseline.

1 Context and Objective

1.1 Why this project

Gigalis operates a central purchasing body for digital services in Pays de la Loire. To anticipate re-tendering workloads, it wants to predict *when* digital contracts come up for renewal. The core obstacle: BOAMP publishes calls for tenders and awards, but *no notice says “this renews contract X”*. The renewal link must be inferred from text, classification, timing, and buyer identity.

1.2 Objective

The internship objective is to build a transparent, reproducible pipeline that (i) reconstructs a proxy renewal **event** from BOAMP signals, (ii) uses it as the outcome of a survival model of contract lifetime, and (iii) translates the model into operational renewal-risk indicators. This

report orients the reader to that pipeline and its main results; exact field logic, formulas, and diagnostics are deferred to the technical and data quality reports.

2 Data Sources and Scope

2.1 Scope

- **Sectors:** CPV divisions 48 (software), 72 (IT services), 32 (telecom), 35 (security).
- **Territory:** departments 44, 49, 53, 72, 85 (Pays de la Loire).
- **Period:** 2015–2024. Study end date: 2024-12-31.
- **Modeling path:** BOAMP only. DECP is documented but not linked (no shared key between the two sources).

2.2 Corpus numbers

- 3,181 BOAMP notices downloaded (2015–2024).
- 1,933 are `APPEL_OFFRE` (source contracts); 1,081 are `ATTRIBUTION`.
- 1,210 are *eligible*: their expected renewal window (± 6 months around the estimated end date) closes before 2024-12-31.
- 723 are structurally censored upfront (too recent to observe a renewal).

DECP (3,039 digital contracts, same scope) is far richer (100% SIRET, 99.4% durations) but cannot be merged to BOAMP at contract level; contract-level DECP merging remains a future path.

2.3 Key data quality issues (modeling consequences)

Table 2: Key BOAMP data-quality findings and their modeling consequence. Full field-level detail is in the data quality report.

Finding	Value	Consequence
Buyer SIRET coverage (raw)	~9% of corpus	Grouping uses normalised names; SIREN enrichment later consolidates many (see §3.2).
Amount fill rate (<code>APPEL_OFFRE</code>)	~21%	Amount excluded from linking signals.
Declared duration granularity	69% 12/24/36/48 mo	at Duration used as a covariate, not as survival target.
Imputed durations	25% of eligible cohort	Flagged with <code>dur_was_imputed</code> ; imputation is 48 months.
No explicit renewal label	structural	<code>event</code> is a proxy reconstructed by the linking algorithm.

2.4 Taxonomy

Each contract is tagged with one of eleven technology categories (Software & Applications, IT Services & Consulting, Telecom & Networks, Cybersecurity, Cloud & Infrastructure, Data & AI, etc.) via a **deterministic rule-based mapping** from CPV codes and description keywords.

This is *not* a trained classifier — the planned NLP classifier was not implemented. Output: `data/processed/taxonomy.csv`.

3 Main Pipeline

The pipeline runs end to end from raw BOAMP notices to operational risk indicators. Table 3 summarises each component and where its outputs live; the subsections below describe what each stage does, and Section 4 reports the numbers.

Table 3: Component status and key outputs. Status: Done, Partial, Not started.

Component	Status	Key result	Where
Data cleaning & EDA	Done	3,181 notices → 1,933 source contracts	data/processed/boamp_full_clean.csv
Buyer SIREN enrichment	Done	724/1,210 eligible rows on a SIREN key	buyer_siren_enrichment/
Renewal linking (event reconstruction)	Done	Selected M2 balanced: 254/1,210 proxy events (21.0%); M0 balanced baseline: 269 (22.2%)	boamp_renewal_linking_quality/, notebooks/06_linkage_method_comparison_no_ground_truth.ipynb
Survival dataset construction	Done	1,210 rows × 33 cols	data/processed/boamp_phase2_survival_method_m2_balanced.csv
Kaplan–Meier & log-rank	Done	Median not reached; fewer than half have observed proxy events before censoring	notebooks/06_linkage_method_comparison_no_ground_truth.ipynb
Cox PH model	Done	C-index 0.553 (M2 balanced, reduced spec; PH-checked)	same notebook + reports/tables/survival/
Parametric AFT models	Done	LogNormalAFT best (AIC 3,357.9 under M2 balanced)	same notebook
Sensitivity analysis	Done	M0 balanced conservative baseline; M0 strict high-confidence sensitivity	reports/tables/validation/method_survival_comparison.csv
Robustness checks	Done	Synthetic benchmark, negative controls, threshold sweep, mapped-audit diagnostic	reports/tables/validation/
Operational risk indicators	Done	1,204 contracts scored; median p12=0.0206, p24=0.0634 (M2 balanced)	reports/tables/survival/renewal_risk_12_24_months_m2_balanced.csv
High-risk contract/buyer/segment tables	Done	Top-20 contracts by 12-month recurrence probability, plus buyer- and segment-level rankings, under the selected M2 balanced method	reports/tables/survival/top20_renewal_risk_m2_balanced.csv, buyer_renewal_risk_ranking_m2_balanced.csv, segment_renewal_risk_ranking_m2_balanced.csv
Manual event validation	Done	150 cases audited; precision ≈0.09–0.15 (HIGH tier 0.50)	event_validation/manual_validation_summary.md
Trained NLP classifier	Partial	Scaffolding only: annotation sample (350 rows) + training scripts exist; gold labels and training not yet done	scripts/nlp*.py, notebooks/03_nlp_classification.ipynb
Change-point detection (Phase 4)	Not done	Scoped but not implemented	—

L3 status (internship-guide deliverable). Mapping the rows above to the internship guide’s deliverable list: **L3, the survival analysis notebook, is Done.** All five required L3 components are present on the selected M2 balanced dataset (with M0 balanced as the conservative baseline sensitivity): Kaplan–Meier curves, a validated Cox model (Section 4 PH check above), a parametric AFT comparison including Weibull (tested, not retained; LogNormalAFT selected by AIC), individual 12- and 24-month predictions, and the high-risk contract/buyer/segment tables. **L2, the trained NLP classifier, is handled separately by the team and does not block L3.** The survival analysis above uses the rule-based CPV/keyword taxonomy (`data/processed/taxonomy.csv`) as a provisional segment covariate; the survival pipeline (notebooks 06–07 and `scripts/task_section16b_calibrated_risk_indicators.py`) can be rerun unchanged once final NLP-classifier labels are delivered.

3.1 BOAMP cleaning

Raw notices are parsed from the BOAMP API, deduplicated, and standardised: buyer key construction, contract start-date refinement (using `ATtribution` back-references where available), duration cleaning and imputation (missing eligible durations set to 48 months, flagged), CPV normalisation, and text normalisation. Output: `data/processed/boamp_full_clean.csv`.

3.2 Buyer SIREN enrichment

Because raw SIRET coverage is weak (~9% of the corpus), buyers were initially grouped by normalised name, which fragments a single public entity across spelling variants. A **SIREN enrichment step** (completed **2026-06-24**) queries the official *API Recherche d’Entreprises* for each buyer name and, for **HIGH-confidence matches only**, upgrades the buyer key from `NAME:` to `SIREN:{9-digit}`, merging name aliases of the same legal entity. Of the 1,210 eligible contracts, **724 carry a SIREN key (plus 2 SIRET) and 484 remain name-keyed**; 275 of 525 unique buyers (52.4%) matched with HIGH confidence. Medium/low matches are kept for review but never applied automatically. Enrichment is a buyer-resolution improvement, *not* ground truth: rare false merges at HIGH confidence remain possible. Outputs: `buyer_siren_enrichment/` and `data/processed/boamp_phase2_survival_siren_enriched.csv`.

3.3 Renewal-event reconstruction

Each eligible source contract is matched against later BOAMP notices from the same buyer using a four-signal composite score: semantic text embeddings (Sentence-Transformer, `paraphrase-multilingual-MiniLM-L12-v2`), CPV hierarchy match, temporal proximity, and buyer-identity agreement (weights 0.40 / 0.25 / 0.20 / 0.15). The best candidate passing the hard text-similarity filter is selected as the renewal successor and sets `event = 1`; contracts with no qualifying successor before study end are right-censored (`event = 0`). See technical report §5 for the full formula.

3.4 Survival modeling

The survival dataset (`data/processed/boamp_phase2_survival.csv`) has one row per eligible contract with observed/censored duration and covariates (`declared_duration_months`, `dur_was_imputed`, `start_year`, category, procedure type). Three model families were fitted in `notebooks/02_survival_modeling_boamp.ipynb`: Kaplan–Meier (with log-rank tests), a Cox proportional-hazards model, and parametric AFT models (Weibull, log-logistic, log-normal). The fitted model was stress-tested with a sensitivity analysis over three event definitions and a battery of robustness checks.

3.5 Operational risk indicators

The best parametric model (log-normal AFT) is applied to all scored contracts to produce conditional renewal probabilities at 12 and 24 months ahead, aggregated to contract, buyer, and segment level. These are prioritisation indicators, not verified renewal-chain probabilities. Output: `reports/tables/survival/renewal_risk_12_24_months_calibrated_balanced.csv`.

4 Main Results

Reading note: the numbers in this section now use the **selected M2 balanced** method, with M0 balanced reported as the conservative baseline. The old 665-event result appears only as a historical pre-calibration baseline and should not be read as the current main result.

4.1 Renewal linking

Table 4: Current proxy-recurrence outcome under the selected M2 balanced method.

Quantity	Value	
Source contracts (APPEL_OFFRE)	1,933	
Eligible (renewal window observable)	1,210	
Proxy recurrence events (<code>event = 1</code>)	254	(21.0%)
Censored/no observed link under method (<code>event = 0</code>)	956	(79.0%)
Conservative baseline (M0 balanced)	269	(22.2%)
Old pre-calibration baseline	665	(55.0%; historical only)

The M0 broad sensitivity gives 490 events (40.5%) and the M0 strict sensitivity gives 79 events (6.5%). M1 balanced (256 events), which never uses the mapped audit labels, closely matches the selected M2 balanced (254 events) and confirms that the method choice does not rest on audit-informed training.

Historical category table. The category table below is retained only to document the pre-calibration baseline and should not be used as the selected method’s event distribution.

Table 5: Historical pre-calibration event rate by technology category (1,210 eligible contracts; events sum to 665).

Category	n	events	rate %
IT Services & Consulting	344	208	60.5
Software & Applications	223	140	62.8
Telecom & Networks	223	116	52.0
Cybersecurity	136	73	53.7
Unknown	115	49	42.6
Digital Workplace & Collaboration	51	32	62.7
Data & AI	40	12	30.0
IT Maintenance & Support	22	11	50.0
IT Hardware & Equipment	26	9	34.6
Cloud & Infrastructure	22	11	50.0
GIS & Mapping	8	4	50.0

Link quality. Of the 665 accepted links: 164 are single-candidate (no competing match); of the remaining 501 multi-candidate links, 219 (44%) have a score margin below 0.05, flagging an

ambiguous winner. 75 links (11.3%) satisfy the **high_confidence_strict** criterion ($S \geq 0.70$ and margin ≥ 0.05 , multi-candidate only); these form the Model C core (Table 8). **Main unlinked failure mode:** 491 of 545 unlinked contracts (90.1%) have no temporal partner in the search window — a true absence of a candidate notice, not a scoring failure.

4.2 Survival analysis

Selected-method survival summary. Under the selected M2 balanced method, the survival dataset has 1,210 rows and 254 proxy events. The Kaplan–Meier median is not reached, meaning fewer than 50% of contracts experience the proxy event before censoring. Fixed-horizon survival is 95.6% at 12 months, 92.1% at 24 months, 84.0% at 48 months, and 76.0% at 60 months. The official reduced-spec Cox C-index is 0.553 (see the two-specification note above; richer category-aware spec 0.606). LogNormalAFT is the best AFT model by AIC (3,357.9), ahead of WeibullAFT (3,404.4) and LogLogisticAFT (3,408.2) — WeibullAFT was tested but not retained since it does not have the lowest AIC. Operational 12/24-month indicators are available under M2 balanced for 1,204 contracts, with median probabilities 0.0206 and 0.0634 and expected counts 29.0 and 84.1. Under the M0 balanced baseline the corresponding results are 269 events, 12-month survival 95.9%, C-index 0.544, LogNormalAFT AIC 3,542.5, and expected counts 27.5 / 82.3 — the headline conclusions are stable across the two methods.

Historical pre-calibration survival. The following detailed tables come from the old 665-event baseline and are retained as provenance for why calibration was needed.

Kaplan–Meier. Global median survival is **50.1 months**, consistent with the prevalence of 48-month framework contracts. Log-rank tests (Table 6): renewal timing is driven by declared duration and imputation status ($p < 0.001$); category is marginally significant ($p = 0.025$).

Table 6: Log-rank tests for differences in renewal timing.

Stratifying variable	statistic	p	df
Category (multivariate)	12.89	0.025	5
Duration was imputed	19.96	< 0.001	1
Declared-duration group	41.05	< 0.001	2
Procedure type (collapsed)	3.48	0.176	2

Cox proportional-hazards model. In-sample concordance: **0.6317**. Two predictors are significant (Table 7): longer declared duration lowers hazard (HR 0.991 per month), and later start year raises it (HR 1.071 per year). IT Services & Consulting shows a positive category effect (HR 1.230, $p = 0.037$). The positive **start_year** hazard is partly structural: 2021–2024 contracts are right-censored because a typical 48-month contract would renew after the 2024-12-31 study close.

Table 7: Cox model: significant covariates (multivariate).

Covariate	HR	95% CI	p
declared_duration_months	0.991	[0.986, 0.995]	< 0.001
start_year	1.071	[1.031, 1.113]	< 0.001
IT Services & Consulting (vs. Cybersecurity)	1.230	[1.013, 1.493]	0.037
dur_was_imputed	0.871	[0.722, 1.049]	0.146
Other category / procedure terms	not significant ($p > 0.1$)		

Parametric AFT models. Log-normal fits best (AIC 7,068.4, concordance 0.655), outperforming log-logistic (7,154.7) and Weibull (7,240.3). Full results: `reports/tables/survival/parametric_aic_comparison.csv`.

Sensitivity to event definition. Three event definitions were tested (Table 8). The Cox concordance drops from 0.632 (Model A) to 0.616 (Model B, ≥ 0.50) and 0.455 (Model C, strict H.C.), suggesting that low-confidence links add predictive signal not present in the strict subset — and that the 75-event strict definition is too sparse for stable estimation.

Table 8: Sensitivity to event definition. “—” = KM median never reached. KM median in months.

Variant	events	rate %	KM median	Cox C
A — baseline (all links)	665	55.0	50.1	0.632
B — conservative (≥ 0.50)	400	33.1	—	0.616
C — strict H.C. ($S \geq 0.70$, $\Delta \geq 0.05$, multi-cand.)	75	6.2	—	0.455

Proportional-hazards assumption. The Schoenfeld test flags a violation for `declared_duration_months` ($p < 10^{-5}$); all other covariates pass. Stratification or a time interaction for duration is future work. The log-normal AFT avoids the PH assumption and gives the same conclusions.

Temporal hold-out. A temporal validation (train on contracts starting ≤ 2021 : 1,088 rows / 604 events; test on ≥ 2022 : 116 rows / 60 events) gives a test concordance of 0.543 against 0.612 in-sample — a directional check only, since the small test cohort is structurally dominated by early renewals, but it confirms the in-sample C-index is an upper bound.

4.3 Operational risk indicators

The table below is computed under the *pre-calibration* baseline event definition (665 events). Under the calibrated balanced rule (269 events; see the calibration update on p. 1), the re-scored indicators are substantially lower: mean 12/24-month probabilities 0.023/0.068, max 12-month probability 0.073, and 27.5 expected recurrences within 12 months (83% lower than the 89.7 below). Files: `reports/tables/survival/*_calibrated_balanced.csv`.

Table 9: Operational risk summary (1,204 scored contracts).

Metric	Value
Mean 12-month renewal probability	0.074
Mean 24-month renewal probability	0.216
Max 12-month probability (any contract)	0.288
Contracts in <i>medium</i> band (0.25–0.50)	6
Contracts in <i>low</i> band (< 0.25)	1,198
Expected renewals within 12 months	89.7
Expected renewals within 24 months	259.9
Top buyer by expected 12-month volume	SIREN:234400034 (9.3)
Top segment by expected 12-month volume	IT Services & Consulting (30.4)

5 Reliability and Limitations

5.1 Reliability evidence

The event variable was stress-tested with checks in `validation_robustness/`:

- **Threshold sensitivity (7 scenarios: baseline, composite floors 0.50–0.90, strict H.C.):** Cox concordance declines gradually from 0.632 (baseline) to 0.599 (score ≥ 0.60), then becomes unstable below ≈ 100 events (0.473 at ≥ 0.70); hazard-ratio directions stay stable at every threshold with enough events. Full table: `validation_robustness/outputs/threshold_sensitivity_summary.csv`.
- **Placebo check:** winning-link mean score 0.554 vs. runner-up 0.428 vs. synthetic no-buyer 0.401 — the scoring discriminates (14.9% of winners reach ≥ 0.70 vs. 0.7% of runner-ups).
- **Event/censoring bias:** SMD = 0.17 (duration), -0.15 (start year) between event and censored groups — covariates reasonably balanced (SMD < 0.20); the dominant bias is structural right-censoring of 2022–2024 contracts.
- **Manual validation (executed 2026-07-02):** 150 stratified cases audited against the full official BOAMP records (`event_validation/outputs/manual_validation_audit_labeled.csv`). Result: TP 14 / FP 82 / uncertain 4 on 100 linked pairs (precision ≈ 0.15 raw, ≈ 0.09 weighted to the 665-event population; 0.50 in the HIGH/strict tier; 1.00 at text similarity ≥ 0.80); 6 of 47 decided unlinked sources hid a plausible missed renewal (12.8%). This is a sample-based audit estimate, not population ground truth; it invalidates the LOW tier and weakens the MEDIUM tier as renewal evidence. Because the sample was stratified on the pre-calibration baseline’s own links and confidence tiers, mapped audit precision is a baseline-anchored diagnostic and is not used to arbitrate between M0, M1, and M2 in the method comparison.

5.2 Key limitations

- **Proxy event, not official ground truth.** Because BOAMP does not provide an official renewal label, the constructed `event` variable should be interpreted as a proxy for likely procurement recurrence, not as verified renewal-chain ground truth — under the initial rule (`event` = 1 means a plausible renewal link was found; by composite score, 39.8% of the 665 links are below 0.50 (low), 45.3% between 0.50 and 0.70 (medium), 14.9% at or above 0.70 (high)) and equally under the calibrated balanced rule.
- **Residual buyer fragmentation.** SIREN enrichment consolidated 724 of the 1,210 eligible rows onto a SIREN key (plus 2 SIRET), but 484 remain name-keyed, so spelling variants not resolved by the API can still create false negatives. Enrichment is not ground truth.
- **Short-text false positives.** Generic descriptions (“prestations informatiques”) can match spuriously.
- **Structural right-censoring.** Contracts from 2021–2024 are largely censored; the positive `start_year` hazard partly reflects study truncation, not a genuine cohort trend.
- **Modest discrimination.** Concordance ≈ 0.65 — useful for prioritisation, not sufficient for high-stakes individual decisions.

6 Completed Work and Remaining Work

Completed. BOAMP acquisition and cleaning; buyer SIREN enrichment (2026-06-24); pre-calibration renewal-event reconstruction (665 events, 55.0%); synthetic benchmark calibration with the real Sentence-Transformer encoder; calibrated balanced event definition (269 events, 22.2%); method comparison M0/M1/M2 with promotion of M2 balanced as the selected main method (254 events, 21.0%; 2026-07-08); survival rerun (KM, Cox with PH check, AFT) under M2 balanced and the M0 sensitivity family; operational 12/24-month risk indicators (pre-calibration, M0 balanced, and M2 balanced versions); rule-based technology taxonomy.

Remaining.

- **Method-neutral audit of the selected method.** The 150-case audit (executed 2026-07-02) showed the broad baseline was too permissive, but its sample is baseline-anchored. The priority is now a method-neutral manual review of the selected M2 balanced links — the 150-pair active-learning queue (`reports/tables/validation/active_learning_review_sample.csv`) is the prepared starting point — to confirm the benchmark-based promotion on real data.
- **Trained NLP classifier (L2).** The technology taxonomy is rule-based (CPV + keywords). The classifier pipeline is scaffolded (`scripts/nlp_annotation_sample.py` through `scripts/nlp_propagate_labels.py` and notebook 03), and the 350-row annotation sample is drawn, but the gold labels have not been filled in and no model has been trained yet. This L2 work is now handled separately by the team and **does not block L3**: the survival analysis uses the available CPV/keyword taxonomy as a provisional segment variable, and the pipeline can be rerun unchanged once final NLP-classifier labels are delivered.
- **Change-point / trend detection (Phase 4).** Scoped but not started.
- **DECP semantic linking.** Applied to BOAMP only; DECP linking remains at the lexical prototype stage.
- **Live prediction scoring.** The current model evaluates contract age as of the study end date; a production run would re-evaluate each contract at a live prediction date.

7 Conclusion and Next Steps

The internship delivered a reproducible end-to-end pipeline: BOAMP notice acquisition → cleaning → buyer SIREN enrichment → renewal-event reconstruction → survival modeling → operational risk indicators. The main methodological result is not a single number but a **change in method**: the project moved from an initial rule that maximized *linkage quantity* (665 events, 55.0%, later shown by manual audit to be only $\approx 9\text{--}15\%$ precise) to calibrated proxy-event definitions selected against a controlled synthetic benchmark and real-data diagnostics — the M0 broad rule gives 490 events (40.5%), the M0 balanced baseline 269 events (22.2%), the M0 strict rule 79 events (6.5%), and the **selected M2 balanced method 254 events (21.0%)**. Under M2 balanced, the Kaplan–Meier median is not reached and LogNormalAFT is the best AFT model by AIC (3,357.9). These outputs translate BOAMP proxy recurrence signals — not externally verified renewal-chain outcomes — into 12/24-month renewal-risk rankings.

Immediate next steps (prioritised):

1. **Run the next survival interpretation on the selected M2 balanced dataset.** Use M0 balanced, broad, and strict only as sensitivity checks; do not report real precision/recall without manual labels.
2. **Re-score contracts at a live prediction date** (current scores use end-of-study ages).

3. **Review medium/low SIREN matches** to extend buyer consolidation beyond the 724 HIGH-confidence rows already applied.
4. **Replace the rule-based taxonomy** with a trained text classifier.
5. **Implement change-point detection** (Phase 4).